

Quick Revision Formula Sheet

Transmission-Line Numericals, Dielectric Boundaries and Electrostatic Energy

General line constants

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}, \quad \gamma = \alpha + j\beta = \sqrt{(R + j\omega L)(G + j\omega C)}.$$

$$v_p = \frac{\omega}{\beta}, \quad \lambda = \frac{2\pi}{\beta} = \frac{v_p}{f}, \quad 1 \text{ Np} = 8.686 \text{ dB}.$$

Lossless line:

$$Z_0 = \sqrt{L/C}, \quad \beta = \omega\sqrt{LC}, \quad C = \frac{\beta}{\omega Z_0}, \quad L = Z_0^2 C.$$

Distortionless line:

$$\frac{R}{L} = \frac{G}{C}, \quad Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{R}{G}}.$$

$$R = \alpha Z_0, \quad G = \frac{\alpha}{Z_0}, \quad L = \frac{Z_0}{v_p}, \quad C = \frac{1}{Z_0 v_p}.$$

Input impedance

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)}.$$

Lossless:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta l)}{Z_0 + jZ_L \tan(\beta l)}.$$

$$Z_{sc} = Z_0 \tanh(\gamma l), \quad Z_{oc} = Z_0 \coth(\gamma l), \quad Z_0 = \sqrt{Z_{oc} Z_{sc}}.$$

Quarter wave: $Z_{in} = Z_0^2 / Z_L$. Half wave: $Z_{in} = Z_L$. For a lossless shorted stub:

$$Z_{in} = jZ_0 \tan \beta l.$$

Reflection and standing waves

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad \Gamma(d) = \Gamma_L e^{-2\gamma d}.$$

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}, \quad Z_{\max} = Z_0 S, \quad Z_{\min} = \frac{Z_0}{S}.$$

For a lossless line, $|\Gamma|$ and S are constant along the line.

Voltage and current waves

With z measured from the input toward the load:

$$V(z) = V^+ e^{-\gamma z} + V^- e^{\gamma z}, \quad I(z) = \frac{V^+}{Z_0} e^{-\gamma z} - \frac{V^-}{Z_0} e^{\gamma z}.$$

$$V^+ = \frac{V_{in} + Z_0 I_{in}}{2}, \quad V^- = \frac{V_{in} - Z_0 I_{in}}{2}.$$

With d measured from the load toward the generator on a lossless line:

$$V(d) = V_0^+ (e^{j\beta d} + \Gamma_L e^{-j\beta d}),$$

$$I(d) = \frac{V_0^+}{Z_0} (e^{j\beta d} - \Gamma_L e^{-j\beta d}).$$

Dielectric boundary, no free surface charge

$$\mathbf{E}_{t1} = \mathbf{E}_{t2}, \quad \mathbf{D}_{n1} = \mathbf{D}_{n2}, \quad \mathbf{D} = \epsilon \mathbf{E}.$$

Thus $E_{n2} = (\epsilon_1/\epsilon_2)E_{n1}$ and $D_{t2} = (\epsilon_2/\epsilon_1)D_{t1}$. With surface charge:

$$\hat{\mathbf{n}} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \rho_s.$$

Electrostatic energy

$$w_e = \frac{1}{2} \mathbf{D} \cdot \mathbf{E} = \frac{1}{2} \epsilon |\mathbf{E}|^2 = \frac{|\mathbf{D}|^2}{2\epsilon}, \quad W_e = \iiint_V w_e dv.$$

Point charges:

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}}.$$

Isolated sphere:

$$C = 4\pi\epsilon a, \quad W = \frac{Q^2}{2C} = \frac{Q^2}{8\pi\epsilon a}.$$

Numerical Problem Types and Variants

Transmission Lines, Dielectric Boundaries and Electrostatic Energy

Revised pass: duplicate questions merged; transmission-line examples on pp. 41–49 of the merged text added. Plane-wave and Smith-chart problems are excluded.

Notation and verification rules used in this revision

Only numerical problems are included. Pure derivations and theory-only questions are omitted. The handwritten author uses nonstandard-looking symbols: in particular, Z_0 can resemble γ , and β , l , 1 , j , μ , and m can be visually similar. Every symbol is therefore interpreted from its formula, units, and surrounding calculation rather than from shape alone. Where a handwritten result conflicts with the stated data and the standard formula, the value is recomputed and the correction is identified.

Contents

1	Transmission-Line Numericals	4
1.1	Type 1: Z_0 , γ , and phase velocity from primary constants	4
1.2	Type 2: Find L and C of a lossless line from Z_0, β, f	4
1.3	Type 3: Distortionless line from Z_0, α, v_p, f	5
1.4	Type 4: Load reflection coefficient and VSWR	6
1.5	Type 5: Reflection coefficient at a distance from the load	6
1.6	Type 6: Lossy-line Z_{in} , sending current and current at an interior point	7
1.7	Type 7: Matched line driven by a source	7
1.8	Type 8: Complete $v(d, t)$ and $i(d, t)$ from generator and load data	7
1.9	Type 9: Input impedance for special electrical lengths and terminations	8
1.9.1	9A. Arbitrary lossless length: $l = 0.35\lambda$	8
1.9.2	9B. Lossless line of length $\lambda/6$	8
1.9.3	9C. Short-circuited lossless line	9
1.9.4	9D. Short-circuited lossy coaxial line	9
1.9.5	9E. Choose a shorted-line length to imitate a capacitor	9
1.9.6	9F. Physical length and frequency given	9
1.10	Type 10: Open-circuit and short-circuit measurements	10
1.10.1	10A. Find Z_0 from magnitudes	10
1.10.2	10B. Find Z_{sc} when Z_0 and Z_{oc} are known	10
1.10.3	10C. Full extraction of Z_0, γ, R, L, G, C	10
1.11	Type 11: Cascaded quarter-wave branches	11
2	Dielectric-Boundary Numericals	11
2.1	Type 12: Dielectric interface with a specified surface charge	11
2.2	Type 13: Vector field crossing a planar dielectric boundary	12
2.3	Type 14: Find $\mathbf{E}_2$, $\mathbf{D}_2$, and refraction angles	12
2.4	Type 15: Boundary problem stated in terms of $\mathbf{D}_1$	13
2.5	Type 16: Radial field through concentric dielectric shells	13
3	Electrostatic-Energy Numericals	13
3.1	Type 17: Energy of several point charges	13
3.2	Type 18: Work required to assemble two charges	14
3.3	Type 19: Integrate energy density in a spherical region	14
3.4	Type 20: Cylindrical potential $V = 80\phi$	14
3.4.1	20A. Full azimuthal region	14

3.4.2	20B. Restricted azimuth and potential difference	14
3.5	Type 21: Energy from a Cartesian potential over a cube	15
3.6	Type 22: Isolated charged sphere	15

1 Transmission-Line Numericals

1.1 Type 1: Z_0 , γ , and phase velocity from primary constants

Question. A line has

$$R = 10 \, \Omega/\text{km}, \quad L = 250 \, \text{mH}/\text{km}, \quad G = 50 \, \text{mS}/\text{km}, \quad C = 100 \, \text{mF}/\text{km}.$$

Find the characteristic impedance at $\omega = 2000 \, \text{rad/s}$.

Procedure

Keep all four distributed constants in the same length unit. Then use

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}.$$

The ratio has units Ω^2 ; choose the square-root branch with positive real part.

$$Z_0 = \sqrt{\frac{10 + j(2000)(0.25)}{0.05 + j(2000)(0.1)}} = \boxed{1.5812 - j0.0156 \, \Omega}.$$

Frequency variants

ω (rad/s)	Rectangular Z_0 (Ω)	Polar Z_0
500	$1.5824 - j0.0624$	$1.5837 \angle -2.258^\circ$
1000	$1.5815 - j0.0312$	$1.5818 \angle -1.131^\circ$
2000	$1.5812 - j0.0156$	$1.5813 \angle -0.566^\circ$

What changed: only ω . Re-evaluate both $R + j\omega L$ and $G + j\omega C$; do not scale a previous answer directly.

CIE variant: find Z_0 , γ , and v_p

A telephone line has

$$R = 30 \, \Omega/\text{km}, \quad L = 100 \, \text{mH}/\text{km}, \quad G = 0, \quad C = 20 \, \mu\text{F}/\text{km}, \quad f = 1 \, \text{kHz}.$$

Keep all constants per kilometre. With $\omega = 2\pi \times 10^3$,

$$Z_0 = \sqrt{\frac{30 + j\omega(0.1)}{j\omega(20 \times 10^{-6})}} = \boxed{70.731 - j1.688 \, \Omega} = 70.751 \angle -1.367^\circ \, \Omega.$$

$$\gamma = \sqrt{(30 + j\omega(0.1))(j\omega(20 \times 10^{-6}))} = \boxed{0.2121 + j8.8883 \, \text{km}^{-1}}.$$

Hence

$$v_p = \frac{\omega}{\beta} = \boxed{706.9 \, \text{km/s} = 7.069 \times 10^5 \, \text{m/s}}.$$

What changed: the question asks for all three outputs. Use the quotient for Z_0 , the product for γ , and ω/β for phase velocity. The printed $20 \, \mu\text{F}/\text{km}$ is used exactly as stated.

Source clarification

The handwritten symbol resembles γ , but the quotient formula and ohmic units identify it as Z_0 . The actual propagation constant uses the product formula. The two lower-frequency handwritten arithmetic lines are inconsistent with the stated constants; the values above are recomputed.

Source: *transmissionl&Boundarycondns.pdf*, p. 1; *Improvement Test question paper*, Q5

1.2 Type 2: Find L and C of a lossless line from Z_0 , β , f

Representative question. A lossless line has $Z_0 = 70 \, \Omega$, $\beta = 3 \, \text{rad/m}$ at $f = 100 \, \text{MHz}$. Find L and C per metre.

Procedure

For a lossless line,

$$Z_0 = \sqrt{L/C}, \quad \beta = \omega\sqrt{LC}.$$

Eliminate L or C :

$$C = \frac{\beta}{\omega Z_0}, \quad L = Z_0^2 C, \quad \omega = 2\pi f.$$

$$C = \frac{3}{2\pi(100 \times 10^6)(70)} = 68.21 \text{ pF/m},$$

$$L = 70^2(68.21 \times 10^{-12}) = \boxed{334.23 \text{ nH/m}}.$$

All distinct variants

Z_0	β	f	C	L
70 Ω	3 rad/m	100 MHz	68.21 pF/m	334.23 nH/m
50 Ω	20 rad/m	700 MHz	90.95 pF/m	227.36 nH/m
50 Ω	5 rad/m	120 MHz	132.63 pF/m	331.57 nH/m
100 Ω	12 rad/m	50 MHz	381.97 pF/m	3.820 μ H/m
30 Ω	10 rad/m	75 MHz	707.36 pF/m	636.62 nH/m

What changed: Z_0 , β , and f . The procedure is unchanged. Convert MHz to Hz before calculating ω .Source: *transmissionl&Boundarycondns.pdf*, pp. 5–6; *Transmission_Lines_merged.pdf*, pp. 41 and 46**1.3 Type 3: Distortionless line from Z_0, α, v_p, f** **Question.** A distortionless line has

$$Z_0 = 60 \Omega, \quad \alpha = 20 \text{ mNp/m}, \quad v_p = 0.6c.$$

Find R, L, G, C , and λ at $f = 100 \text{ MHz}$.**Procedure**

For a distortionless line,

$$\frac{R}{L} = \frac{G}{C}, \quad Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{R}{G}}, \quad \alpha = \sqrt{RG}.$$

The fastest working formulas are

$$R = \alpha Z_0, \quad G = \frac{\alpha}{Z_0}, \quad L = \frac{Z_0}{v_p}, \quad C = \frac{1}{Z_0 v_p}, \quad \lambda = \frac{v_p}{f}.$$

Convert mNp to Np first.

With $\alpha = 0.020 \text{ Np/m}$ and $v_p = 1.8 \times 10^8 \text{ m/s}$,

$$\boxed{R = 1.20 \Omega/\text{m}}, \quad \boxed{G = 333.33 \mu\text{S/m}},$$

$$\boxed{L = 333.33 \text{ nH/m}}, \quad \boxed{C = 92.59 \text{ pF/m}},$$

$$\beta = \frac{\omega}{v_p} = 3.4907 \text{ rad/m}, \quad \boxed{\lambda = 1.80 \text{ m}}.$$

What may be changedIf Z_0, α , or v_p changes, use the same five direct formulas. Frequency affects β and λ , but not R, L, G, C when the stated distortionless parameters are treated as frequency-independent.Source: *Transmission_Lines_merged.pdf*, p. 46

1.4 Type 4: Load reflection coefficient and VSWR

Representative question. A line has $Z_0 = 25 \Omega$ and $Z_L = 60 + j40 \Omega$. Find Γ_L and VSWR.

Procedure

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad S = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}.$$

Line length and frequency do not affect Γ_L at the load; they matter only when moving the coefficient to another point.

$$\Gamma_L = \frac{35 + j40}{85 + j40} = 0.5184 + j0.2266 = 0.5658 \angle 23.61^\circ,$$

$$S = \frac{1 + 0.5658}{1 - 0.5658} = \boxed{3.606}.$$

All distinct load variants

$Z_0 (\Omega)$	$Z_L (\Omega)$	Γ_L	VSWR
25	$60 + j40$	$0.5184 + j0.2266$	3.606
50	$40 + j25$	$-0.0315 + j0.2865$	1.810
30	$50 + j40$	$0.4000 + j0.3000$	3.000
50	$100 + j50$	$0.4000 + j0.2000$	2.618
50	$25 + j36$	$-0.0837 + j0.5202$	3.227

What changed: only Z_0 and Z_L . Complex division is the only non-routine step; calculate $|\Gamma|$ before VSWR.

Frequency-dependent series- RC load

For $Z_0 = 100 \Omega$, $R_L = 50 \Omega$, $C_L = 10 \text{ pF}$, and $f = 100 \text{ MHz}$,

$$Z_L = R_L + \frac{1}{j\omega C_L} = 50 - j159.155 \Omega.$$

Then

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = 0.37278 - j0.66550 = \boxed{0.76279 \angle -60.744^\circ}.$$

If VSWR is also required, $S = 7.431$. **Procedure change:** form the frequency-dependent load impedance first; the reflection formula itself is unchanged.

Source: *transmissionl&Boundarycondns.pdf*, pp. 2–3; *Transmission_Lines_merged.pdf*, pp. 41–42

1.5 Type 5: Reflection coefficient at a distance from the load

Question. A lossless line has $Z_0 = 50 \Omega$, $Z_L = 100 \Omega$. Find the reflection coefficient a distance $d = \lambda/4$ toward the generator.

$$\Gamma_L = \frac{100 - 50}{100 + 50} = \frac{1}{3}.$$

For a lossless line,

$$\Gamma(d) = \Gamma_L e^{-j2\beta d}, \quad \beta = \frac{2\pi}{\lambda}.$$

Thus

$$2\beta d = 2 \left(\frac{2\pi}{\lambda} \right) \frac{\lambda}{4} = \pi, \quad \Gamma(\lambda/4) = \frac{1}{3} e^{-j\pi} = \boxed{-\frac{1}{3} = \frac{1}{3} \angle 180^\circ}.$$

Procedure for another distance

Replace $\lambda/4$ by the new distance. On a lossless line the magnitude stays $|\Gamma_L|$; only the phase changes by $-2\beta d$.

Source: *Zoandreflectioncoeffproblems.pdf*, p. 8

1.6 Type 6: Lossy-line Z_{in} , sending current and current at an interior point

Question. A line has $\alpha = 8$ dB/m, $\beta = 1$ rad/m, $Z_0 = 60 + j40 \Omega$, $l = 2$ m, $Z_L = 20 + j50 \Omega$. It is driven by $V_g = 10\angle 0^\circ$ V through $Z_g = 40 \Omega$. Find Z_{in} , I_{in} , and the current at the midpoint.

Procedure

1. Convert attenuation: $\alpha_{\text{Np}} = \alpha_{\text{dB}}/8.686$.
2. Use the lossy input-impedance formula.
3. Apply the source divider: $I_{in} = V_g/(Z_g + Z_{in})$, $V_{in} = I_{in}Z_{in}$.
4. Obtain V^+ and V^- , then evaluate $I(z)$ at $z = l/2$.

$$\alpha = \frac{8}{8.686} = 0.9210 \text{ Np/m}, \quad \gamma = 0.9210 + j1.$$

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh(\gamma l)}{Z_0 + Z_L \tanh(\gamma l)} = \boxed{60.25 + j38.79 \Omega}.$$

$$I_{in} = \frac{10}{40 + Z_{in}} = \boxed{93.03\angle -21.15^\circ \text{ mA}}.$$

$$V_{in} = I_{in}Z_{in}, \quad V^+ = \frac{V_{in} + Z_0 I_{in}}{2}, \quad V^- = \frac{V_{in} - Z_0 I_{in}}{2}.$$

At $z = 1$ m,

$$I(1) = \frac{V^+}{Z_0} e^{-\gamma} - \frac{V^-}{Z_0} e^{\gamma} = \boxed{34.92\angle 280.92^\circ \text{ mA}}.$$

Source: Zoandreflectioncpeffproblems.pdf, pp. 1–2; Transmission_Lines_merged.pdf, pp. 47–48

1.7 Type 7: Matched line driven by a source

Question. A 40 m line is matched: $Z_L = Z_0 = 30 + j60 \Omega$. The source is $V_g = 15\angle 0^\circ$ V_{rms}, $Z_g = 30 + j60 \Omega$. Find Z_{in} , I_{in} , and V_{in} .

$$Z_{in} = Z_0 = \boxed{30 + j60 \Omega}.$$

Since $Z_g = Z_{in}$,

$$V_{in} = \frac{V_g Z_{in}}{Z_g + Z_{in}} = \frac{V_g}{2} = \boxed{7.5\angle 0^\circ \text{ V}_{\text{rms}}},$$

$$I_{in} = \frac{V_g}{2Z_0} = \boxed{0.1118\angle -63.435^\circ \text{ A}}.$$

Arithmetic check

The handwritten current magnitude appears smaller. Direct division by $2(30+j60)$ gives 0.1118 A; the voltage-divider result remains 7.5 V.

Source: Zoandreflectioncpeffproblems.pdf, p. 3

1.8 Type 8: Complete $v(d, t)$ and $i(d, t)$ from generator and load data

Question. A 50Ω , 0.67 m lossless line at 1.05 GHz has $v_p = 0.7c$, $Z_L = 100 + j50 \Omega$, $Z_g = 10 \Omega$, and

$$v_g(t) = 10 \sin(\omega t + 30^\circ) \text{ V}.$$

Find the voltage and current everywhere on the line. Let d be distance from the load toward the generator.

Procedure

1. Convert the source to the cosine-reference phasor.
2. Find λ, β, Γ_L , and Z_{in} .
3. Use the source divider to obtain V_{in} .
4. Solve for the incident-wave amplitude V_0^+ at the load reference.
5. Substitute into the lossless d -coordinate wave equations.

$$\begin{aligned}\tilde{V}_g &= 10\angle -60^\circ \text{ V}, & \lambda &= \frac{0.7c}{1.05 \times 10^9} = 0.20 \text{ m}, \\ \beta &= 10\pi \text{ rad/m}, & \beta l &= 126^\circ \pmod{360^\circ}, \\ \Gamma_L &= \frac{100 + j50 - 50}{100 + j50 + 50} = 0.4 + j0.2 = 0.44721\angle 26.565^\circ.\end{aligned}$$

At $d = l$,

$$Z_{in} = Z_0 \frac{e^{j\beta l} + \Gamma_L e^{-j\beta l}}{e^{j\beta l} - \Gamma_L e^{-j\beta l}} = \boxed{21.886 + j17.433 \Omega}.$$

$$V_{in} = \tilde{V}_g \frac{Z_{in}}{Z_g + Z_{in}} = 7.6996\angle -50.128^\circ \text{ V},$$

$$V_0^+ = \frac{V_{in}}{e^{j\beta l} + \Gamma_L e^{-j\beta l}} = 10.1773\angle 158.965^\circ \text{ V}.$$

Therefore

$$v(d, t) = 10.177 \cos(\omega t + \beta d + 158.965^\circ) + 4.551 \cos(\omega t - \beta d - 174.470^\circ) \text{ V},$$

$$i(d, t) = 0.20355 \cos(\omega t + \beta d + 158.965^\circ) - 0.09103 \cos(\omega t - \beta d - 174.470^\circ) \text{ A}.$$

The current may equivalently be written with a plus sign and reflected-wave phase 5.530° .

Reflected-current sign

The reflected current has the opposite sign to the reflected voltage:

$$I(d) = \frac{V_0^+}{Z_0} (e^{j\beta d} - \Gamma_L e^{-j\beta d}).$$

A printed solution that uses a plus sign with the same reflected-voltage phase misses this sign reversal.

Source: *Transmission_Lines_merged.pdf*, pp. 42–44

1.9 Type 9: Input impedance for special electrical lengths and terminations

9A. Arbitrary lossless length: $l = 0.35\lambda$

Given $Z_0 = 100 \Omega$, $Z_L = 60 + j30 \Omega$:

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = -0.20755 + j0.22642 = 0.30715\angle 132.51^\circ,$$

$$S = \frac{1 + 0.30715}{1 - 0.30715} = \boxed{1.887}.$$

With $\beta l = 2\pi(0.35) = 0.7\pi$,

$$Z_{in} = 100 \frac{(60 + j30) + j100 \tan(0.7\pi)}{100 + j(60 + j30) \tan(0.7\pi)} = \boxed{64.84 - j38.28 \Omega}.$$

Source: *tmlinesandqs.pdf*, p. 20; duplicated in *cie3solution.pdf*, p. 8

9B. Lossless line of length $\lambda/6$

Given $Z_0 = 50 \Omega$, $Z_L = 120 \Omega$:

$$\Gamma_L = 0.41176, \quad S = 2.4,$$

$$\beta l = \frac{\pi}{3}, \quad Z_{in} = \boxed{26.26 - j22.55 \Omega = 34.61\angle -40.66^\circ \Omega}.$$

Variation procedure: calculate $\beta l = 2\pi l/\lambda$, then use the lossless input-impedance formula.

Source: *Zoandreflectionncpeffproblems.pdf*, p. 16

9C. Short-circuited lossless line

Given $Z_0 = 75 \, \Omega$, $l = 1 \, \text{cm}$, $f = 3 \, \text{GHz}$, and $v_p = c$:

$$\beta l = \frac{2\pi f}{c} l = 0.2\pi = 36^\circ,$$

$$Z_{in} = jZ_0 \tan(\beta l) = \boxed{j54.49 \, \Omega}.$$

The input is **inductive** under the standard current and distance convention.

Sign convention

The handwritten sheet marks the result as capacitive using a minus sign. Earlier formulas in the same notes use $Z_{sc} = jZ_0 \tan \beta l$. The standard result for $0 < l < \lambda/4$ is inductive.

Source: *Zoandreflectioncoeffproblems.pdf*, p. 4

9D. Short-circuited lossy coaxial line

Given $Z_0 = 65 + j38 \, \Omega$, $\gamma = 0.7 + j2.25 \, \text{m}^{-1}$, $l = 0.8 \, \text{m}$:

$$Z_{in} = Z_0 \tanh(\gamma l) = \boxed{132.47 + j29.13 \, \Omega}.$$

Recomputed hyperbolic function

The handwritten sinh and cosh values do not correspond to $\gamma l = 0.56 + j1.8$, so its boxed answer is not retained. Evaluate $\tanh(0.56 + j1.8)$ directly in rectangular complex form.

Source: *Zoandreflectioncoeffproblems.pdf*, p. 15

9E. Choose a shorted-line length to imitate a capacitor

A shorted $50 \, \Omega$ lossless line at $2.25 \, \text{GHz}$ must have the same input impedance as $C_{eq} = 4 \, \text{pF}$. The wave velocity is $0.75c$.

For the shorted line and capacitor,

$$jZ_0 \tan \beta l = -\frac{j}{\omega C_{eq}}, \quad \tan \beta l = -\frac{1}{Z_0 \omega C_{eq}} = -0.35368.$$

Also

$$\beta = \frac{\omega}{v_p} = 62.832 \, \text{rad/m}, \quad \lambda = \frac{v_p}{f} = 0.10 \, \text{m}.$$

The smallest positive solution is in quadrant II:

$$\beta l = \pi - \tan^{-1}(0.35368) = 2.8016,$$

$$\boxed{l_{\min} = 4.459 \, \text{cm}}.$$

All solutions are

$$\boxed{l = 4.459 \, \text{cm} + n \frac{\lambda}{2} = 4.459 \, \text{cm} + 5n \, \text{cm}, \quad n = 0, 1, 2, \dots}$$

Variation procedure: equate $jZ_0 \tan \beta l$ to the required reactance and choose the tangent quadrant that gives the desired inductive or capacitive sign.

Source: *Transmission_Lines_merged.pdf*, pp. 44–45

9F. Physical length and frequency given

A $50 \, \Omega$ lossless line is $30 \, \text{m}$ long, operates at $2 \, \text{MHz}$, has $v_p = 0.6c$, and is terminated by $Z_L = 60 + j40 \, \Omega$.

$$\Gamma_L = \frac{10 + j40}{110 + j40} = 0.35230 \angle 56.31^\circ, \quad S = \boxed{2.088}.$$

$$\beta l = \frac{2\pi f l}{v_p} = 120^\circ.$$

$$Z_{in} = 50 \frac{60 + j40 + j50 \tan 120^\circ}{50 + j(60 + j40) \tan 120^\circ} = \boxed{23.97 + j1.35 \, \Omega}.$$

What changed: the electrical length is not supplied directly. Compute $\beta l = 2\pi fl/v_p$ before using the same lossless-line formula.

Source: *Transmission_Lines_merged.pdf*, pp. 48–49

1.10 Type 10: Open-circuit and short-circuit measurements

10A. Find Z_0 from magnitudes

If $|Z_{oc}| = 100 \, \Omega$ and $|Z_{sc}| = 25 \, \Omega$,

$$Z_0 = \sqrt{Z_{oc}Z_{sc}} = \sqrt{100 \times 25} = \boxed{50 \, \Omega}.$$

Source: *Zoandreflectioncoeffproblems.pdf*, p. 4

10B. Find Z_{sc} when Z_0 and Z_{oc} are known

$$Z_{sc} = \frac{Z_0^2}{Z_{oc}}.$$

Z_0	Z_{oc}	Z_{sc}
50 Ω	100 + j150 Ω	7.692 – j11.538 Ω
75 Ω	50 + j70 Ω	38.007 – j53.209 Ω
30 Ω	25 + j40 Ω	10.112 – j16.180 Ω
60 Ω	25 + j35 Ω	48.649 – j68.108 Ω

Variation procedure: square the real scalar Z_0 , then divide by the complex Z_{oc} using its conjugate.

Source: *transmissionl&Boundarycondns.pdf*, p. 8

10C. Full extraction of Z_0, γ, R, L, G, C

A 500 km line at $\omega = 10^4$ rad/s has

$$Z_{oc} = 2000 \angle -80^\circ \, \Omega, \quad Z_{sc} = 20 \angle 20^\circ \, \Omega.$$

First,

$$Z_0 = \sqrt{Z_{oc}Z_{sc}} = \boxed{200 \angle -30^\circ \, \Omega}.$$

For a short circuit, $Z_{sc} = Z_0 \tanh(\gamma l)$, hence

$$\tanh(\gamma l) = \frac{Z_{sc}}{Z_0} = 0.1 \angle 50^\circ.$$

Using the principal branch,

$$\gamma l = \operatorname{atanh}(0.1 \angle 50^\circ) = 0.063989 + j0.076769,$$

$$\gamma = \boxed{(1.2798 + j1.5354) \times 10^{-7} \, \text{m}^{-1}}.$$

Thus

$$\alpha = 1.2798 \times 10^{-7} \, \text{Np/m}, \quad \beta = 1.5354 \times 10^{-7} \, \text{rad/m}.$$

Use

$$R + j\omega L = \gamma Z_0, \quad G + j\omega C = \frac{\gamma}{Z_0}.$$

The primary constants are

$$\begin{aligned} R &= 3.752 \times 10^{-5} \, \Omega/\text{m}, & L &= 1.380 \times 10^{-9} \, \text{H/m}, \\ G &= 1.703 \times 10^{-10} \, \text{S/m}, & C &= 9.848 \times 10^{-14} \, \text{F/m}. \end{aligned}$$

Branch choice

Because $\tanh(z)$ is periodic in the imaginary direction, β can differ by integer multiples of π/l . Choose the branch consistent with the known frequency, phase velocity, or expected line parameters. The sheet uses the principal branch.

Source: *tmlinesandqs.pdf*, pp. 13–14; duplicated in *cie3solution.pdf*, pp. 1–2

1.11 Type 11: Cascaded quarter-wave branches

Representative question. A $\lambda/4$ main line with $Z_{0m} = 50 \Omega$ feeds two parallel $\lambda/4$ branches. Each branch has $Z_{0b} = 100 \Omega$ and is terminated in 50Ω . Find the impedance seen by the source.

Procedure

1. Transform each branch load: $Z'_k = Z_{0bk}^2 / Z_{Lk}$.
2. Combine the branch inputs in parallel: $Z_J = Z'_1 \parallel Z'_2$.
3. Transform the junction through the main quarter-wave section: $Z_{in} = Z_{0m}^2 / Z_J$.

$$Z'_1 = Z'_2 = \frac{100^2}{50} = 200 \Omega, \quad Z_J = 200 \parallel 200 = 100 \Omega,$$

$$Z_{in} = \frac{50^2}{100} = \boxed{25 \Omega}.$$

All branch-network variants

Main Z_{0m}	Branch Z_{0b}	Each Z_L	Source sees Z_{in}
50 Ω	100 Ω	50 Ω	25 Ω
25 Ω	50 Ω	30 Ω	15 Ω
60 Ω	150 Ω	80 Ω	25.6 Ω
100 Ω	200 Ω	75 Ω	37.5 Ω

What changed: the three impedances. Always transform branches first, combine them, and transform once more through the main section.

Source: *extraQs.pdf*, pp. 1–2

2 Dielectric-Boundary Numericals

2.1 Type 12: Dielectric interface with a specified surface charge

At an interface between $\epsilon_{r1} = 2$ and $\epsilon_{r2} = 4$, let

$$\mathbf{E}_1 = 2\hat{\mathbf{x}} + 4\hat{\mathbf{y}} + 6\hat{\mathbf{z}} \text{ V/m}, \quad \rho_s = 4\epsilon_0 \text{ C/m}^2,$$

with $\hat{\mathbf{z}}$ normal to the interface. Find $\mathbf{E}_2$.

Tangential electric field is continuous:

$$E_{2x} = 2, \quad E_{2y} = 4.$$

Using the sheet's normal orientation,

$$D_{2n} = D_{1n} + \rho_s = 2\epsilon_0(6) + 4\epsilon_0 = 16\epsilon_0,$$

$$E_{2n} = \frac{16\epsilon_0}{4\epsilon_0} = 4.$$

Therefore

$$\boxed{\mathbf{E}_2 = 2\hat{\mathbf{x}} + 4\hat{\mathbf{y}} + 4\hat{\mathbf{z}} \text{ V/m}}.$$

Sign check

Write the boundary equation with an explicit unit normal: $\hat{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \rho_s$. Reversing $\hat{n}$ reverses the algebraic sign of ρ_s , not the physics.

Source: *transmission\&Boundarycondns.pdf*, p. 11

2.2 Type 13: Vector field crossing a planar dielectric boundary

For $z \geq 0$, $\epsilon_{r1} = 4$; for $z \leq 0$, $\epsilon_{r2} = 3$. In region 1,

$$\mathbf{E}_1 = 5\hat{x} - 2\hat{y} + 3\hat{z} \text{ kV/m}.$$

Find $\mathbf{E}_2$, the angles with the interface, the energy density in each region, and the energy in a cube of side 2 m centred at $(3, 4, -5)$.

Procedure

The interface normal is $\hat{z}$. Keep E_x, E_y unchanged. With no free surface charge, $\epsilon_1 E_{1n} = \epsilon_2 E_{2n}$. For an angle with the interface, use $\alpha = \tan^{-1}(|E_n|/|E_t|)$.

$$\mathbf{E}_2 = 5\hat{x} - 2\hat{y} + \frac{4}{3}(3)\hat{z} = \boxed{5\hat{x} - 2\hat{y} + 4\hat{z} \text{ kV/m}}.$$

Since $|E_t| = \sqrt{29} \text{ kV/m}$,

$$\alpha_1 = \tan^{-1} \frac{3}{\sqrt{29}} = \boxed{29.12^\circ}, \quad \alpha_2 = \tan^{-1} \frac{4}{\sqrt{29}} = \boxed{36.62^\circ}.$$

$$w_{e1} = \frac{1}{2}(4\epsilon_0)(38 \times 10^6) = \boxed{672.92 \text{ } \mu\text{J/m}^3},$$

$$w_{e2} = \frac{1}{2}(3\epsilon_0)(45 \times 10^6) = \boxed{597.66 \text{ } \mu\text{J/m}^3}.$$

The cube spans $-6 \leq z \leq -4$, so it lies entirely in region 2. Its volume is 8 m^3 :

$$W = w_{e2}(8) = \boxed{4.781 \text{ mJ}}.$$

Source: *transmissionl&Boundarycondns.pdf*, pp. 22–23; shorter duplicate on p. 18

2.3 Type 14: Find $\mathbf{E}_2$, $\mathbf{D}_2$, and refraction angles

The plane $x = 0$ separates $\epsilon_{r1} = 3$ for $x > 0$ from $\epsilon_{r2} = 5$ for $x < 0$. Given

$$\mathbf{E}_1 = 80\hat{x} - 60\hat{y} - 30\hat{z} \text{ V/m},$$

find the fields in region 2 and the angles with the normal.

The normal is $\hat{x}$. Thus

$$E_{2x} = \frac{\epsilon_1}{\epsilon_2} E_{1x} = \frac{3}{5}(80) = 48,$$

while the tangential components remain -60 and -30 :

$$\boxed{\mathbf{E}_2 = 48\hat{x} - 60\hat{y} - 30\hat{z} \text{ V/m}}.$$

$$\mathbf{D}_1 = (2.125\hat{x} - 1.594\hat{y} - 0.797\hat{z}) \text{ nC/m}^2,$$

$$\boxed{\mathbf{D}_2 = (2.125\hat{x} - 2.656\hat{y} - 1.328\hat{z}) \text{ nC/m}^2}.$$

Angles with the normal:

$$\theta_1 = \tan^{-1} \frac{\sqrt{60^2 + 30^2}}{80} = \boxed{39.99^\circ},$$

$$\theta_2 = \tan^{-1} \frac{\sqrt{60^2 + 30^2}}{48} = \boxed{54.40^\circ}.$$

Source: *transmissionl&Boundarycondns.pdf*, p. 19

2.4 Type 15: Boundary problem stated in terms of $\mathbf{D}_1$

Region 1 has $\epsilon_{r1} = 2$, region 2 has $\epsilon_{r2} = 5$, and the interface normal is $\hat{z}$. Given

$$\mathbf{D}_1 = 3\hat{x} - 4\hat{y} + 5\hat{z} \text{ nC/m}^2,$$

find $\mathbf{D}_2$, its normal and tangential parts, energy densities, its angle with $\hat{z}$, and $|D_2|/|D_1|$.

Normal D is continuous; tangential E is continuous:

$$\mathbf{D}_{2n} = 5\hat{z} \text{ nC/m}^2,$$

$$\mathbf{D}_{2t} = \frac{\epsilon_2}{\epsilon_1} \mathbf{D}_{1t} = \frac{5}{2}(3\hat{x} - 4\hat{y}) = 7.5\hat{x} - 10\hat{y}.$$

Therefore

$$\begin{aligned} \mathbf{D}_2 &= 7.5\hat{x} - 10\hat{y} + 5\hat{z} \text{ nC/m}^2. \\ w_{e1} &= \frac{|D_1|^2}{2\epsilon_1} = \boxed{1.412 \text{ } \mu\text{J/m}^3}, \quad w_{e2} = \frac{|D_2|^2}{2\epsilon_2} = \boxed{2.047 \text{ } \mu\text{J/m}^3}. \\ \theta &= \cos^{-1} \frac{5}{\sqrt{7.5^2 + 10^2 + 5^2}} = \boxed{68.20^\circ}, \\ \frac{|D_2|}{|D_1|} &= \frac{\sqrt{181.25}}{\sqrt{50}} = \boxed{1.904}. \end{aligned}$$

Source: *transmissionl&Boundarycondns.pdf*, p. 20

2.5 Type 16: Radial field through concentric dielectric shells

Two spherical surfaces are at $r = 4$ cm and $r = 9$ cm. The shell $4 < r < 6$ cm has $\epsilon_{r1} = 2$; the shell $6 < r < 9$ cm has $\epsilon_{r2} = 5$. In region 1,

$$\mathbf{E}_1 = \frac{2000}{r^2} \hat{r} \text{ V/m}.$$

Find $\mathbf{E}_2$ and the energy in each shell.

Because the field is normal to the interface and there is no free surface charge,

$$\epsilon_1 E_1 = \epsilon_2 E_2, \quad \mathbf{E}_2 = \frac{800}{r^2} \hat{r} \text{ V/m}.$$

Using $dv = 4\pi r^2 dr$:

$$\begin{aligned} W_1 &= \frac{1}{2}(2\epsilon_0) \int_{0.04}^{0.06} \left(\frac{2000}{r^2} \right)^2 4\pi r^2 dr = \boxed{3.709 \text{ mJ}}, \\ W_2 &= \frac{1}{2}(5\epsilon_0) \int_{0.06}^{0.09} \left(\frac{800}{r^2} \right)^2 4\pi r^2 dr = \boxed{0.989 \text{ mJ}}. \end{aligned}$$

Source: *transmissionl&Boundarycondns.pdf*, p. 21

3 Electrostatic-Energy Numericals

3.1 Type 17: Energy of several point charges

Three charges $-1 \text{ } \mu\text{C}$, $4 \text{ } \mu\text{C}$, and $3 \text{ } \mu\text{C}$ are at $(0, 0, 0)$, $(0, 0, 1)$, and $(1, 0, 0)$, respectively. Find the stored energy.

The pair distances are $r_{12} = 1$, $r_{13} = 1$, $r_{23} = \sqrt{2}$ m:

$$W = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{1} + \frac{q_1 q_3}{1} + \frac{q_2 q_3}{\sqrt{2}} \right) = \boxed{13.35 \text{ mJ}}.$$

Unit interpretation

The handwritten charge symbol resembles “mC”, but the written energy scale is consistent with μC . Literal millicoulomb values would produce kilojoules.

Source: *transmissionl&Boundarycondns.pdf*, p. 14

3.2 Type 18: Work required to assemble two charges

Bring $q_1 = 1 \mu\text{C}$ to $(-2, 6, 1)$, then $q_2 = -2 \mu\text{C}$ to $(3, -4, 0)$. The separation is

$$r_{12} = \sqrt{(5)^2 + (-10)^2 + (-1)^2} = \sqrt{126} \text{ m}.$$

The work is the second charge times the potential created by the first:

$$W = q_2 \frac{q_1}{4\pi\epsilon_0 r_{12}} = \boxed{-1.601 \text{ mJ}}.$$

A negative result means the field releases energy during assembly.

Source: *transmissionl&Boundarycondns.pdf*, p. 14

3.3 Type 19: Integrate energy density in a spherical region

Given

$$\mathbf{E} = 2r \sin \theta \cos \phi \hat{r} + r \cos \theta \cos \phi \hat{\theta} - r \sin \phi \hat{\phi},$$

find the energy for $0 \leq r \leq 2$, $0 \leq \theta \leq \pi$, $0 \leq \phi \leq 2\pi$.

$$|E|^2 = r^2 (4 \sin^2 \theta \cos^2 \phi + \cos^2 \theta \cos^2 \phi + \sin^2 \phi),$$

$$W = \frac{\epsilon_0}{2} \int_0^2 \int_0^\pi \int_0^{2\pi} |E|^2 r^2 \sin \theta d\phi d\theta dr.$$

The dimensionless integral is $256\pi/5$, so

$$W = \frac{128\pi\epsilon_0}{5} = \boxed{0.7121 \text{ nJ}}.$$

If “hemispherical” was intended literally

The written limits describe a full sphere. For either symmetric half-sphere, the energy is half: 0.3560 nJ.

Source: *transmissionl&Boundarycondns.pdf*, pp. 14–15

3.4 Type 20: Cylindrical potential $V = 80\phi$

$$\mathbf{E} = -\nabla V = -\frac{80}{\rho} \hat{\phi}, \quad w_e = \frac{1}{2} \epsilon_0 \frac{6400}{\rho^2}.$$

20A. Full azimuthal region

For $2 < \rho < 4 \text{ cm}$, $0 < \phi < 2\pi$, $0 < z < 1 \text{ m}$:

$$W = \frac{1}{2} \epsilon_0 (6400) \int_0^1 dz \int_0^{2\pi} d\phi \int_{0.02}^{0.04} \frac{1}{\rho^2} \rho d\rho = \boxed{123.45 \text{ nJ}}.$$

Maximum energy density occurs at the smallest ρ :

$$w_{\max} = \frac{1}{2} \epsilon_0 \frac{6400}{(0.02)^2} = \boxed{70.83 \mu\text{J/m}^3}.$$

20B. Restricted azimuth and potential difference

For the same radial and axial limits but $0 < \phi < 0.2\pi$:

$$W = \boxed{12.345 \text{ nJ}}.$$

For $A(3 \text{ cm}, 0, 0)$ and $B(3 \text{ cm}, 0.2\pi, 1 \text{ m})$:

$$V_{AB} = V_A - V_B = 80(0) - 80(0.2\pi) = \boxed{-50.27 \text{ V}}.$$

What changed: only the azimuthal span; energy scales linearly with $\Delta\phi$. The maximum density is unchanged because the radial limits are unchanged.

Source: *transmissionl&Boundarycondns.pdf*, pp. 15 and 26

3.5 Type 21: Energy from a Cartesian potential over a cube

In free space,

$$V = \frac{20}{xyz} \text{ V},$$

for the cube $1 < x, y, z < 2$. Find total energy and the volume of a uniform-energy region whose density equals the value at the cube centre.

$$\mathbf{E} = -\nabla V = 20 \left(\frac{\hat{x}}{x^2 y z} + \frac{\hat{y}}{x y^2 z} + \frac{\hat{z}}{x y z^2} \right).$$

$$W = \frac{1}{2} \epsilon_0 \iiint_1^2 |E|^2 dx dy dz = \boxed{387.37 \text{ pJ}}.$$

At $(1.5, 1.5, 1.5)$,

$$w_c = \frac{1}{2} \epsilon_0 |E_c|^2 = 207.29 \text{ pJ/m}^3.$$

Thus the required uniform volume is

$$V_{eq} = \frac{W}{w_c} = \boxed{1.869 \text{ m}^3}.$$

Source: *transmissionl&Boundarycondns.pdf*, pp. 24–25

3.6 Type 22: Isolated charged sphere

A conducting sphere of radius $a = 4 \text{ cm}$ carries $Q = 5 \text{ } \mu\text{C}$. Find $\mathbf{D}$ outside, total field energy, and capacitance.

$$\mathbf{D}(r) = \frac{5 \times 10^{-6}}{4\pi r^2} \hat{r} \text{ C/m}^2, \quad r \geq 0.04 \text{ m}.$$

$$W = \frac{Q^2}{8\pi\epsilon_0 a} = \boxed{2.809 \text{ J}}.$$

$$C = 4\pi\epsilon_0 a = \boxed{4.451 \text{ pF}}.$$

Source: *transmissionl&Boundarycondns.pdf*, p. 25

Final 2-Minute Formula Sheet

General line

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}},$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}.$$

$$v_p = \omega / \beta, \quad \lambda = 2\pi / \beta, \quad \alpha_{\text{dB}} = \alpha_{\text{dB}} / 8.686.$$

Lossless line

$$Z_0 = \sqrt{L/C}, \quad \beta = \omega \sqrt{LC},$$

$$C = \frac{\beta}{\omega Z_0}, \quad L = Z_0^2 C.$$

Distortionless line

$$R = \alpha Z_0, \quad G = \alpha / Z_0,$$

$$L = Z_0 / v_p, \quad C = 1 / (Z_0 v_p).$$

Input impedance

$$Z_{in} = Z_0 \frac{Z_L + Z_0 \tanh \gamma l}{Z_0 + Z_L \tanh \gamma l}.$$

Lossless: $\tanh \gamma l \rightarrow j \tan \beta l$.

$$Z_{sc} = Z_0 \tanh \gamma l, \quad Z_{oc} = Z_0 \coth \gamma l.$$

$$Z_0 = \sqrt{Z_{oc} Z_{sc}}, \quad \tanh \gamma l = Z_{sc} / Z_0.$$

Quarter wave: $Z_{in} = Z_0^2 / Z_L$.

Reflection

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}, \quad \Gamma(d) = \Gamma_L e^{-2\gamma d}.$$

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}.$$

Waves, input coordinate

$$V^\pm = \frac{V_{in} \pm Z_0 I_{in}}{2}.$$

$$V(z) = V^+ e^{-\gamma z} + V^- e^{\gamma z},$$

$$I(z) = \frac{V^+}{Z_0} e^{-\gamma z} - \frac{V^-}{Z_0} e^{\gamma z}.$$

Waves, load-distance coordinate

$$V(d) = V_0^+ (e^{j\beta d} + \Gamma_L e^{-j\beta d}),$$

$$I(d) = \frac{V_0^+}{Z_0} (e^{j\beta d} - \Gamma_L e^{-j\beta d}).$$

Shorted lossless stub

$$Z_{in} = j Z_0 \tan \beta l.$$

To imitate C :

$$\tan \beta l = -\frac{1}{\omega C Z_0}.$$

Dielectric boundary No free surface charge:

$$\mathbf{E}_{t1} = \mathbf{E}_{t2}, \quad \mathbf{D}_{n1} = \mathbf{D}_{n2}.$$

$$E_{n2} = \frac{\epsilon_1}{\epsilon_2} E_{n1}, \quad D_{t2} = \frac{\epsilon_2}{\epsilon_1} D_{t1}.$$

With surface charge:

$$\hat{n} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \rho_s.$$

Energy density

$$w_e = \frac{1}{2} \epsilon E^2 = \frac{D^2}{2\epsilon}, \quad W_e = \iiint w_e dv.$$

$$dv_{\text{cyl}} = \rho d\rho d\phi dz, \quad dv_{\text{sph}} = r^2 \sin \theta dr d\theta d\phi.$$

Point charges and sphere

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i < j} \frac{q_i q_j}{r_{ij}}.$$

$$C_{\text{sphere}} = 4\pi\epsilon a, \quad W_{\text{sphere}} = \frac{Q^2}{8\pi\epsilon a}.$$

Fast checks

- Quotient under the root $\rightarrow Z_0$; product $\rightarrow \gamma$.
- Matched load: $\Gamma = 0$, $S = 1$, $Z_{in} = Z_0$.
- Open/short: $|\Gamma| = 1$.
- Lossless $\lambda/4$: impedance inversion.
- Reflected current carries a minus sign.
- Energy density cannot be negative.